

KMSG Capstone Project

Carnegie Mellon University – Heinz College

USER GUIDE: Nonprofit Funder Discovery Pipeline: Stage 1: final_kmsg_data.ipynb

1. Purpose of This Guide

This guide explains how to:

- Use the pipeline as-is to generate the dataset
- Understand what each section of the code is doing
- Make changes (e.g., apply it to another state or adjust criteria)
- Safely modify the pipeline without breaking results

This is designed to be clear, step-by-step, and usable at any level of technical background, while still allowing for future expansion.

2. What This Pipeline Does

This pipeline takes raw public data and turns it into a clean dataset of funding relationships.

Specifically, it:

1. Finds nonprofit organizations
2. Identifies which ones are large funders
3. Pulls their official IRS filings
4. Extracts who they gave money to
5. Cleans and organizes that data into a usable format

The final result is a dataset showing: “Who gave money, to whom, how much, and for what purpose.”

3. How to Use the Pipeline (As-Is)

a. Step 1: Open the File in:

- i. Google Colab (recommended)
- ii. Jupyter Notebook
- iii. Any Python notebook environment

b. Step 2: Run All Cells in Order from top to bottom

- i. The pipeline is designed so each step builds on the previous one
- ii. Do not skip sections and Do not run out of order

c. Step 3: Wait for Processing

- i. Some sections take longer because they:
 1. Download IRS data
 2. Call APIs
 3. Parse XML files
- ii. This is expected. The code includes:

1. Progress updates
 2. Checkpoints to prevent data loss
- d. **Step 4: Locate the Output**
- i. The final dataset will be saved as: **final_clean_kmsg_project_df.csv**
 1. *This file is your main output and can be used immediately*

4. How the Pipeline Is Structured

The pipeline is divided into sections. Each section has a specific job.

- a. **Section 1: Setup and Configuration**
 - i. **What it does:** Defines key settings for the entire pipeline.
 - ii. **Why it matters:** This controls what data is collected and how the pipeline behaves.
 - iii. **Key things you can change here:**
 1. State (currently Pennsylvania)
 2. Years of data
 3. Minimum revenue threshold
- b. **Section 2: Helper Functions**
 - i. **What it does:** Cleans and standardizes data (names, EINs, ZIP codes).
 - ii. **Why it matters:** Ensures data matches correctly across sources.
 - iii. **Impact of changes:** Changing this can affect matching accuracy.
- c. **Section 3: Load Initial Nonprofit Data**
 - i. **What it does:** Loads all nonprofits from IRS data.
 - ii. **Why it matters:** This is the starting pool of organizations.
- d. **Section 4: Revenue Filtering**
 - i. **What it does:** Keeps only organizations above a certain revenue level.
 - ii. **Why it matters:** Focuses on organizations that are likely funders.
 - iii. **Key setting:** MIN_REVENUE = 1_000_000
 - iv. **If you change this:**
 1. Lower value → more organizations, more noise
 2. Higher value → fewer, larger funders
- e. **Section 5: Final Funder List**
 - i. **What it does:** Creates the official list of funders used in the pipeline.
- f. **Section 6: ProPublica Enrichment**
 - i. **What it does:** Adds additional data (financials, classifications).
- g. **Section 7: IRS XML Extraction**
 - i. **What it does:** Pulls detailed data from IRS filings (mission, website, etc.).
- h. **Section 8: Data Merging**
 - i. **What it does:** Combines all organization-level data into one table.

- i. **Section 9: Grant Extraction**
 - i. **What it does:** Extracts actual grant records (this is the most important step).
 - ii. **Output:** One row per grant.
 - j. **Section 10: Add Funder Details**
 - i. **What it does:** Adds funder information to each grant.
 - k. **Section 11: Reference Data Matching**
 - i. **What it does:** Matches organizations to a clean reference dataset.
 - ii. **Why it matters:** Improves accuracy and fills missing information.
 - l. **Section 12: Final Cleaning**
 - i. **What it does:** Cleans and standardizes all data into final format.
 - m. **Section 13: Final Data Recovery**
 - i. **What it does:** Fills remaining missing values where possible.
 - n. **Section 14: Final Export**
 - i. **What it does:** Saves the final dataset.
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5. How to Modify the Pipeline

This section explains **exactly what to change and what will happen**.

- a. **Change the State (Most Important Use Case)**
 - i. **Step 1: Update State Variable**
 - 1. Find this in Section 1: `TARGET_STATE = "PA"`
 - 2. Change to: `TARGET_STATE = "CA"` # Example: California
 - ii. **Step 2: Update IRS Dataset**
 - 1. Find: `eo_pa.csv`
 - 2. Replace with: `eo_ca.csv`
 - iii. **What This Changes**
 - 1. Entire dataset will now reflect the new state
 - 2. All funders and recipients will be based on that state
 - iv. **Important Note:** If no reference dataset exists for the new state:
 - 1. The pipeline will still run
 - 2. But matching quality may decrease
- b. **Change Revenue Threshold**
 - i. **Find:** `MIN_REVENUE = 1_000_000`
 - ii. **Effects:**
 - 1. Lower value → more organizations, more noise
 - 2. Higher value → fewer, larger funders
- c. **Change Time Period**
 - i. **Find:** `TARGET_YEARS = list(range(2020, 2024))`

- ii. **Effects**
 - 1. Expanding years → more data, longer runtime
 - 2. Reducing years → faster, less data
 - d. **Adjust Matching Behavior** (Matching happens in Section 11)
 - i. **Current logic:**
 - 1. Match by EIN (strong)
 - 2. If missing → match by name (fallback)
 - ii. **If you change this:**
 - 1. More aggressive matching → more matches, lower accuracy
 - 2. More strict matching → fewer matches, higher accuracy
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6. Common Use Cases

- a. **Find Funders for a Nonprofit**
 - i. Filter dataset by:
 - 1. Cause area
 - 2. Location
 - b. **Identify Largest Funders**
 - i. Sort by: Total grant amount
 - c. **Analyze Funding Gaps**
 - i. Look for:
 - 1. Areas with low funding
 - 2. Underrepresented cause areas
 - d. **Build a Tool or Dashboard**
 - i. Use dataset in: Tableau, Power BI, Python applications
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7. Troubleshooting

- a. **Issue: Pipeline stops mid-run**
 - i. **Solution:** Restart from last checkpoint (automatically saved)
 - b. **Issue: Missing data**
 - i. **Explanation:** Some nonprofits do not report complete data
 - c. **Issue: Slow performance**
 - i. **Explanation:** XML extraction is the most time-intensive step
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8. Summary

This pipeline converts complex public data into a clear, structured dataset of funding relationships.